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Two-Fraction Proton Beam Stereotactic Radiosurgery for Cerebral Arteriovenous Malformations

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Purpose/Objective(s): Stereotactic radiosurgery (SRS) is a well-established treatment option that can obliterate cerebral arteriovenous malformations (AVMs). However, larger AVMs are more challenging to manage, with lower rates of obliteration and higher risk of complications. In this study, we report the obliteration rate and associated adverse effects of planned two-fraction proton SRS (PSRS) used to treat patients with larger AVMs that were deemed inoperable.

Materials/Methods: A retrospective analysis including 75 patients with cerebral AVMs who received two-fraction PSRS at our institution between 1991 and 2019 was conducted. The median AVM nidus volume was 21.5 cc (range 1.4-58.1 cc). AVMs were located in critical/deep locations (basal ganglia, thalamus, or brainstem) in 21.3%. The median total dose for PSRS was 16 Gy(RBE) delivered in two fractions at a median of 7 days apart (range 1-82 days). Factors associated with obliteration and hemorrhage were assessed via univariable and multivariable analyses.

Results: Median follow up was 90 months (range 0.9-223.4 months). Total (8 patients, 11%) or partial (28 patients, 37%) obliteration was achieved among 48% of AVMs at a median time of 85.4 months (range 12.2-222.7 months). Five-year cumulative incidence of total obliteration was 29%. Five-year cumulative incidence of hemorrhage post-PSRS was 33%. Lesions with higher AVM scores were more likely to hemorrhage ($P = 0.042$). The most common adverse effect was Grade 1 headache acutely (11%) and long-term (9%). One patient developed a Grade 2 generalized seizure disorder.

Conclusion: Planned two-fraction PSRS may be an effective and safe treatment option for achieving obliteration in patients with larger cerebral AVMs that are inoperable. Post-treatment hemorrhage remains a potential adverse event.

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Should Adjuvant Radiotherapy be Used in Atypical Meningioma (WHO Grade II) Following Gross Total Resection? A Systematic Review and Meta-Analysis

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Purpose/Objective(s): The role of adjuvant radiotherapy (RT) following gross total resection (GTR) in atypical meningioma (AM) is not well established. Non-randomized prospective trials, such as RTOG 0539, suggest to treat this group with adjuvant RT. However, it remains unclear if there is any improvement in progression free survival (PFS) and overall survival (OS) with this approach. We aim to evaluate the survival benefit of adjuvant RT in AM following GTR.

Materials/Methods: We performed a systematic review and meta-analysis to aggregate the clinical outcomes of patients with AM who had undergone GTR, treated with and without adjuvant RT. We searched biomedical databases, such as Medline, EMBASE and CENTRAL, for studies published between January 1964 and February 2021. Both prospective and retrospective studies which reported the primary outcomes of 5-year PFS and 5-year OS were included. Extracted data was analyzed using random-effects meta-analysis of time to event using the DerSimonian and Laird methods.

Results: Thirteen studies, involving 1124 patients, were included in our meta-analysis. 832 (74.0%) patients were treated with GTR alone and 302 (26.9%) patients had GTR followed by adjuvant RT. Adjuvant RT was delivered as external beam RT (3D conformal RT, intensity modulated RT, fractionated stereotactic radiotherapy) in 10 (76.9%) studies. Three (23.1%) studies also included patients treated with stereotactic radiosurgery in addition to fractionated RT. Median RT dose was 59.4 Gy. Adjuvant RT resulted in an increase of 3.9 months for restricted mean PFS truncated at 60 months (95% confidence interval (CI) 0.23-7.72; $\boldsymbol{P} = 0.037$) and a non-statistically significant 22% reduction in the hazard of disease progression or death (hazards ratio 0.78; 95% CI 0.46-1.33; $\boldsymbol{P} = 0.370$). Restricted mean OS, truncated at 60 months, was improved with adjuvant RT by 1.1 months (95% CI 0.37-1.81; $\boldsymbol{P} = 0.003$) and a non-significant 21% reduction in the hazard of death from any cause (HR 0.79; 95% CI 0.51-1.24; $\boldsymbol{P} = 0.310$).

Conclusion: The use of adjuvant RT in patients with AM who have undergone GTR improved restricted mean PFS and OS. While we await the results from ongoing randomized controlled trials, adjuvant RT should be recommended.

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Clinical Characteristics and Post-Operative Complications as Predictors of Radiation Toxicity After Treatment With I^{125} Eye Plaque Brachytherapy for Uveal Melanomas

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Purpose/Objective(s): I^{125} Eye Plaque brachytherapy is the standard treatment for medium size uveal melanomas. As a result, most patients have late radiation toxicities (RTT) such as radiation-induced maculopathy (RM), radiation-induced neovascular glaucoma/iris neovascularization (RNGI) and radiation-induced optic neuropathy (RON) that result in vision loss. While tumor size, location and radiation dose have been identified as predictors of radiation toxicity, we hypothesize demographics, pre-treatment tumor characteristics and post-operative complications may also impact radiation toxicities.

Materials/Methods: An IRB-approved retrospective chart review was performed from 2011-2019 to evaluate patients with posterior uveal melanomas who were treated with brachytherapy at a single institution in the Southeast with a large Hispanic population, with a minimum of 3 months follow-up. We collected patient demographics, pre-treatment tumor characteristics, tumor margin to fovea/optic disk and post-operative complications as potential predictors of RTT. RTT was defined as an event of RM, RNGI, or RON following plaque removal. Univariable analysis (UVA) and multivariable analysis (MVA) were performed using logistic regression model to examine the associations between study variables and the development of RTT. Odds ratios (ORs) and corresponding P-values were reported. All tests were two-sided and statistical significance was considered when $P < 0.05$. SAS version 9.4 and R version 4.0.8 3 were used for statistical analysis.

Results: A total of 262 patients were evaluated over 8 years. Median follow up was 33.5 months (range 3.02 − 97.31). 178 pts (67.9%) had RTT. 131 pts (50.0%) developed RM. 19 pts (7.3%) developed RNGI. 56 pts (21.4%) developed RO. 41 pts (15.8%) identified as Hispanic. UVA found shorter distance to fovea (OR = 0.97, $P = 0.025$), post-operative subretinal fluid/exudative retinal detachment (RD) (OR = 2.89, $P = 0.002$) and vitreous hemorrhage (OR = 3.73, $P = 0.002$) are associated with RTT and Hispanic ethnicity is not significantly associated with RTT when compared to non-Hispanics (OR = 1.54,